Introduction & Overview

Aim 1: Feature Engineering & Relative Performance

Aim 2: Model Detection Performance by Days Observed

Aim 3: The Cost of High Sensitivity Screening

Wearables & Multiple Sclerosis

DATASCI 223

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Project Aims

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Aim 1

Create day-level features from step count data and evaluate their performance.

Aim 2

Establish how many days of data are needed for reliable patient-level classification.

Aim 3

Establish under realistic MS prevalence (1:400) screening tradeoffs at high sensitivity.

Datasets

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BarKA-MS

Dataset of 44 individuals with multiple sclerosis involved in the Barrieren zu Körperlicher Aktivität in Multipler Sklerose (BarKA-MS) trial.

FitBit Fitness Tracker

Dataset of 32 respondents to a distributed survey via Amazon Mechanical Turk between 03.12.2016-05.12.2016.

LifeSnaps

Dataset of 71 individuals over 4 months involved in the European H2020 Real-time Analytics for Internet of Sports (RAIS) project.

Data Overview

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- Three public Fitbit datasets merged
- Each dataset stripped to ID, datetime, and steps only
- Each participant capped at 720 hours (30 days)
- 147 total participants (44 MS, 103 healthy)

Source	Class	Participants	Hours	Participant %
barka	MS	44	31680	29.9
fitbit	Healthy	32	23040	21.8
lifesnaps	Healthy	71	51120	48.3
Total	-	147	105840	100.0

Figure: Merged Dataset Summary Table

Feature Distribution

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- Derived 13 day-level features
- Clear right-skew pattern across variables, which motivated the use of robust scaling.
- Similar looking distributions motivated correlation analysis

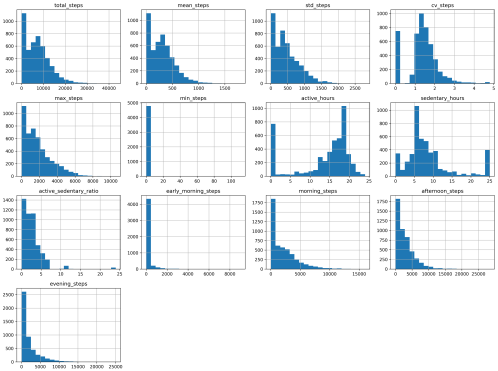

Figure: Feature Distribution Plot

Correlation Plot Part 1.

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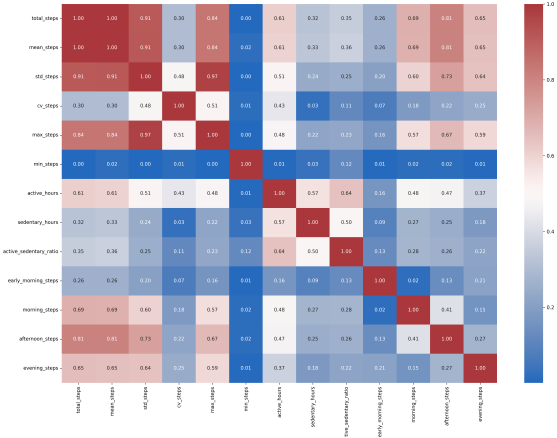

Correlation Plot Part 2.

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- Highly correlated features inflate coefficient variance and produce unreliable odds ratios. In addition, it inflates dimensionality without providing additional meaningful information.
- The following six features were removed: mean_steps, std_steps, max_steps, min_steps, sedentary_hours, active_sedentary _ratio.
- Final set of features: total_steps, cv_steps, active_hours, early_morning_steps, morning_steps, afternoon_steps, evening_steps

Model Results & Feature Importance

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	fold	auc	precision	recall	f1
0	1	0.823010	0.554987	0.777778	0.647761
1	2	0.847377	0.500000	0.826613	0.623100
2	3	0.883807	0.627586	0.880645	0.732886
3	4	0.871695	0.523342	0.858871	0.650382
4	5	0.794971	0.482906	0.810036	0.605087
5	Mean	0.844172	0.537764	0.830789	0.651843

Figure: K-Fold Result

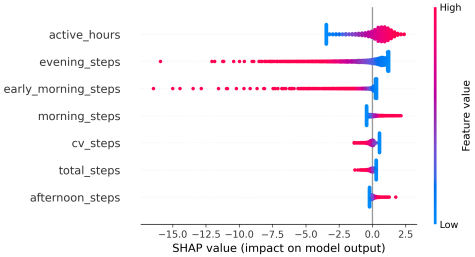

Figure: Feature Importance Plot

Patient-Level AUC by Days Observed

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- AUC stabilizes around 0.93 between days 5-7
- Minimal gains after day 7 ($\approx +0.02$)
- Suggests 1 week is sufficient for stable classification

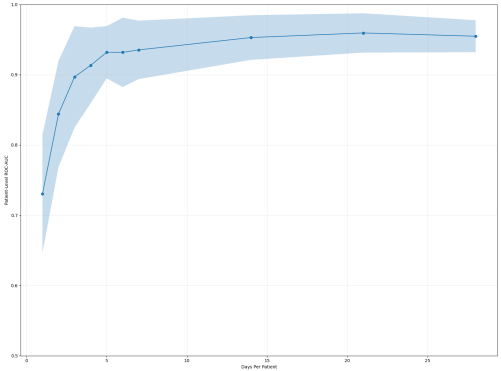

Figure: Patient-Level AUC by Days Observed

Screening Tradeoffs at High Sensitivity Part 1.

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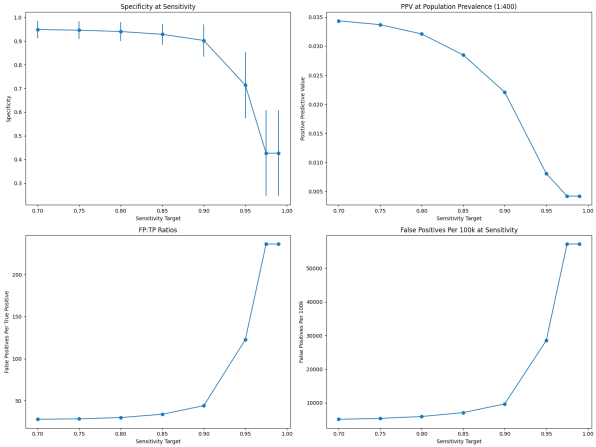

Screening Tradeoffs at High Sensitivity Part 2.

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- 85% Sensitivity: 7100 FP per 100k; 34 FP per TP
- 95% Sensitivity: 28500 FP per 100k; 123 FP per TP
- The current model is insufficient for clinical application and would result in a huge referral burden. A larger dataset is needed.

	sensitivity_target	mean_sensitivity	mean_specificity	std_specificity	mean_TP_100k	mean_FP_100k	mean_Referral_100k	mean_PPV	fp_tp_ratio
0	0.700	0.729343	0.948627	0.036939	182.335710	5124.445788	5306.781497	0.034359	28.104455
1	0.750	0.751640	0.945987	0.037999	187.910072	5387.840011	5575.750083	0.033701	28.672439
2	0.800	0.788724	0.940400	0.040042	197.180937	5945.144679	6142.325616	0.032102	30.150707
3	0.850	0.834479	0.928690	0.044197	208.619644	7113.174706	7321.794350	0.028493	34.096380
4	0.900	0.876435	0.902777	0.068558	219.108833	9697.964272	9917.073104	0.022094	44.260946
5	0.950	0.930987	0.713764	0.138632	232.746791	28552.002465	28784.749257	0.008086	122.674097
6	0.975	0.968478	0.426682	0.181432	242.119603	57188.495733	57430.615335	0.004216	236.199362
7	0.990	0.968478	0.426682	0.181432	242.119603	57188.495733	57430.615335	0.004216	236.199362

Figure: Screening Results Table